Cloud Assignment - Lab6

Name: Harissh Ragav Dhamodaran
Roll no: 2023BCD0059

1)

train_and_export_model.py

```
import json
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression

# Load dataset
data = fetch_california_housing()

X = data.data
y = data.target

# Train model
model = LinearRegression()
model.fit(X, y)

# Save weights
model_data = {
    "coefficients": model.coef_.tolist(),
    "intercept": model.intercept_
}

with open("model.json", "w") as f:
    json.dump(model_data, f)

print("Model saved")
```

Used to train the linear regression model and export model weights.

server.js

```
const express = require('express')
const fs = require('fs')

const app = express()
app.use(express.json())

// Load model
```

```
const model = JSON.parse(fs.readFileSync('./model.json'))

const weights = model.coefficients
const intercept = model.intercept

// prediction function
function predict(features) {

   let result = intercept

   for (let i = 0; i < weights.length; i++) {
       result += weights[i] * features[i]
   }

   return result
}

// API endpoint
app.post('/predict', (req, res) => {

   const features = req.body.features

   if (!features || features.length !== weights.length) {
       return res.status(400).send("Invalid input")
   }

   const prediction = predict(features)

   res.json({
       prediction: prediction
   })

})

app.listen(3000, () => {
   console.log("Server running on port 3000")
})
```

Dockerfile

```
FROM node:18-alpine

WORKDIR /app
```

```
COPY package*.json ./

RUN npm install --only=production

COPY server.js .
COPY model.json .

EXPOSE 3000

CMD ["node", "server.js"]
```

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment

metadata:
 name: housing-deployment

spec:
 replicas: 1

 selector:
   matchLabels:
     app: housing-service

 template:
   metadata:
     labels:
       app: housing-service

   spec:
     containers:
       - name: housing-container
         image: housing-service:latest
         imagePullPolicy: Never

         ports:
           - containerPort: 3000
```

service.yaml

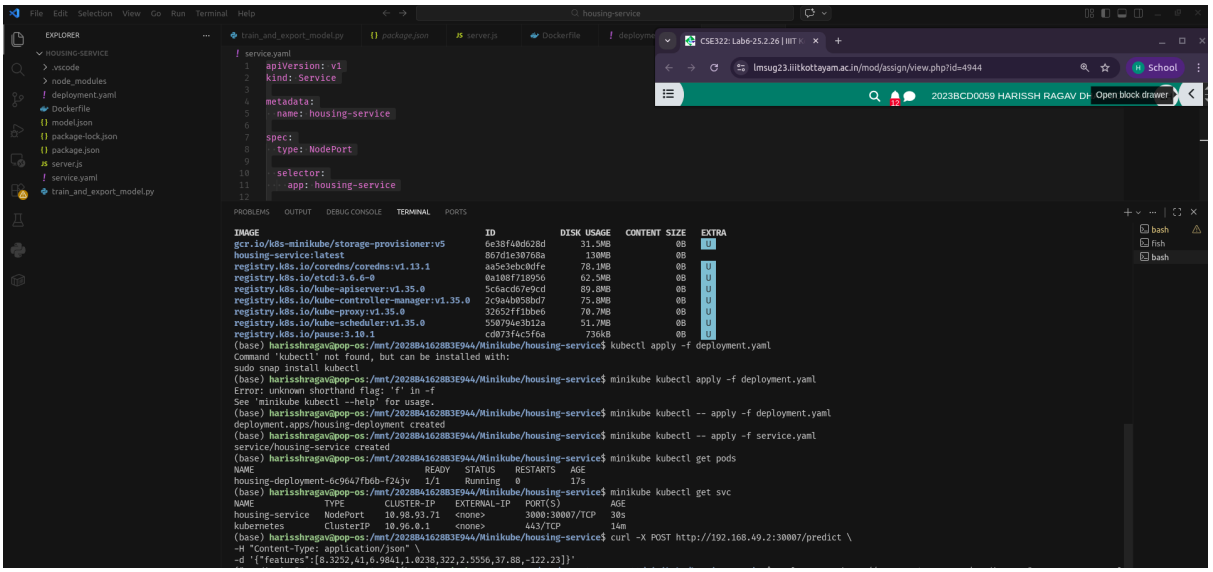
```yaml
apiVersion: v1
kind: Service
```

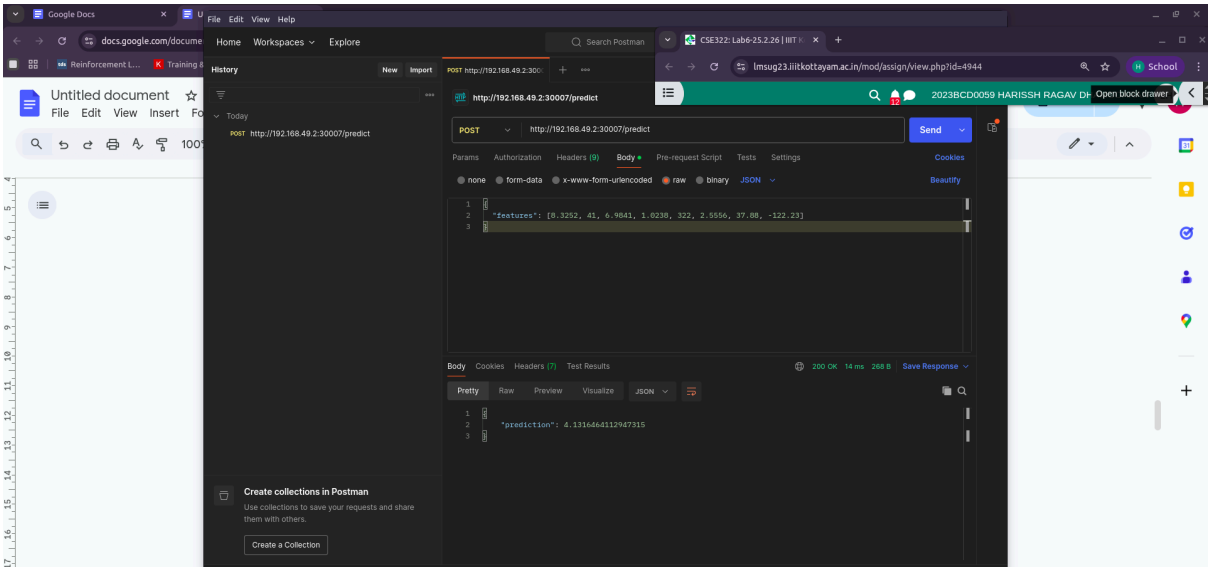
```yaml
metadata:
 name: housing-service

spec:
 type: NodePort

 selector:
   app: housing-service

 ports:
   - protocol: TCP
     port: 3000
     targetPort: 3000
     nodePort: 30007
```




Google Docs
Untitled document
File Edit View Insert
Home Workspaces Explore
History
Today
POST http://192.168.49.2:30007/predict
POST http://192.168.49.2:30007/predict
Send
Body Cookies Headers (7) Test Results
200 OK 14 ms 268 B Save Response
Pretty Raw Preview Visualize JSON
"prediction": 4.1316464112947315
Create collections in Postman
Use collections to save your requests and share them with others.
Create a Collection
lmsug23.iiitkottayam.ac.in/mod/assign/view.php?id=4944
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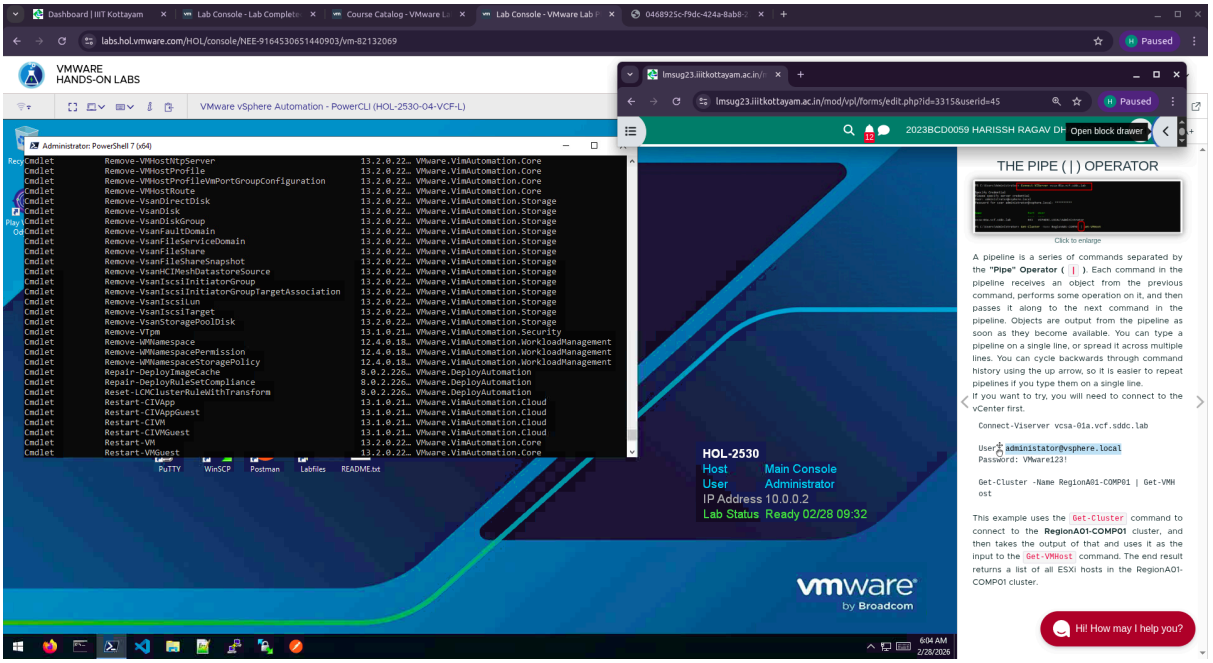
2)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VMWARE HANDS-ON LABS
VMware vSphere Automation - PowerCLI (HOL-2530-04-VCF-L)
Administrator: PowerShell 7 (x64)
HOL-2530
Host Main Console
User Administrator
IP Address 10.0.0.2
Lab Status Ready 02/28 09:32
vmware by Broadcom
lmsug23.iiitkottayam.ac.in/mod/vpl/forms/edit.php?id=3315&userid=45
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THE PIPE (|) OPERATOR
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User: administrator@vsphere.local
Password: VMware123!
Get-Cluster -Name RegionA01-COMP01 | Get-VMHost
This example uses the Get-Cluster command to connect to the RegionA01-COMP01 cluster, and then takes the output of that and uses it as the input to the Get-VMHost command. The end result returns a list of all ESXi hosts in the RegionA01-COMP01 cluster.
Hi! How may I help you?

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labs.hol.vmware.com/HOL/console/NEE-91645306514409 03/vm-82132069

VMWARE HANDS-ON LABS

VMware vSphere Automation - PowerCLI (HOL-2530-04-VCF-L)

Administrator: PowerShell 7 (x64)

```
Cmdlet          Test-VMHostProfileCompliance                13.2.0.22… VMware.VimAutomation.Core
Cmdlet          Test-VMHostSnmp                             13.2.0.22… VMware.VimAutomation.Core
Cmdlet          Test-VsanClusterHealth                      13.2.0.22… VMware.VimAutomation.Storage
Cmdlet          Test-VsanNetworkPerformance                 13.2.0.22… VMware.VimAutomation.Storage
Cmdlet          Test-VsanStoragePerformance                 13.2.0.22… VMware.VimAutomation.Storage
Cmdlet          Test-VsanVMCreation                         13.2.0.22… VMware.VimAutomation.Storage
Cmdlet          Uninstall-HCXSentinel                       13.0.0.20… VMware.VimAutomation.Hcx
Cmdlet          Unlock-VM                                   13.1.0.21… VMware.VimAutomation.Security
Cmdlet          Unregister-KeyProvider                      13.1.0.21… VMware.VimAutomation.Security
Cmdlet          Update-Entity                               12.7.0.20… VMware.VumAutomation
Cmdlet          Update-HCXSentinel                          13.0.0.20… VMware.VimAutomation.Hcx
Cmdlet          Update-Tools                                13.2.0.22… VMware.VimAutomation.Core
Cmdlet          Update-VsanHclDatabase                      13.2.0.22… VMware.VimAutomation.Storage
Cmdlet          Use-PowerCLIContext                         13.2.0.22… VMware.VimAutomation.Common
Cmdlet          Wait-ApplianceBackupJob                     13.2.0.22… VMware.VimAutomation.Core
Cmdlet          Wait-HCXJob                                 13.0.0.20… VMware.VimAutomation.Hcx
Cmdlet          Wait-Task                                   13.2.0.22… VMware.VimAutomation.Common
Cmdlet          Wait-Tools                                  13.2.0.22… VMware.VimAutomation.Core
Cmdlet          Wait-VcfTask                                13.3.0.24… VMware.Sdk.Vcf.SddcManager
Cmdlet          Wait-VcfValidation                          13.3.0.24… VMware.Sdk.Vcf.SddcManager

PS C:\Users\Administrator> GET-VMHost

Name                  ConnectionState PowerState NumCpu CpuUsageMhz CpuTotalMhz   MemoryUsageGB   MemoryTotalGB Version
----                  --------------- ---------- ------ ----------- -----------   -------------   ------------- -------
esx-02a.vcf.sddc.lab Connected       PoweredOn       2         277        4198           1.951           5.999   8.0.3
esx-03a.vcf.sddc.lab Connected       PoweredOn       2         303        4198           2.119           5.999   8.0.3
esx-01a.vcf.sddc.lab Connected       PoweredOn       2         146        4198           1.920           5.999   8.0.3

PS C:\Users\Administrator>
```

PuTTY WinSCP Postman Labfiles README.txt

HOL-2530
Host Main Console
User Administrator
IP Address 10.0.0.2
Lab Status Ready 02/28 09:32

vmware by Broadcom

6:05 AM 2/28/2026

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PS C:\Users\Administrator> (Get-VM).Count
6
PS C:\Users\Administrator>  $esxihost = Get-VMHost
PS C:\Users\Administrator> ($esxihost | Select -First
>> 1).ConnectionState.ToString()
ParserError:
Line |
   1 |  ($esxihost | Select -First
     |                           ~
     | Missing closing ')' in expression.
PS C:\Users\Administrator> (
PS C:\Users\Administrator>
PS C:\Users\Administrator>
PS C:\Users\Administrator> ($esxihost | Select -First 1).ConnectionState.ToString()
Connected
PS C:\Users\Administrator>
```

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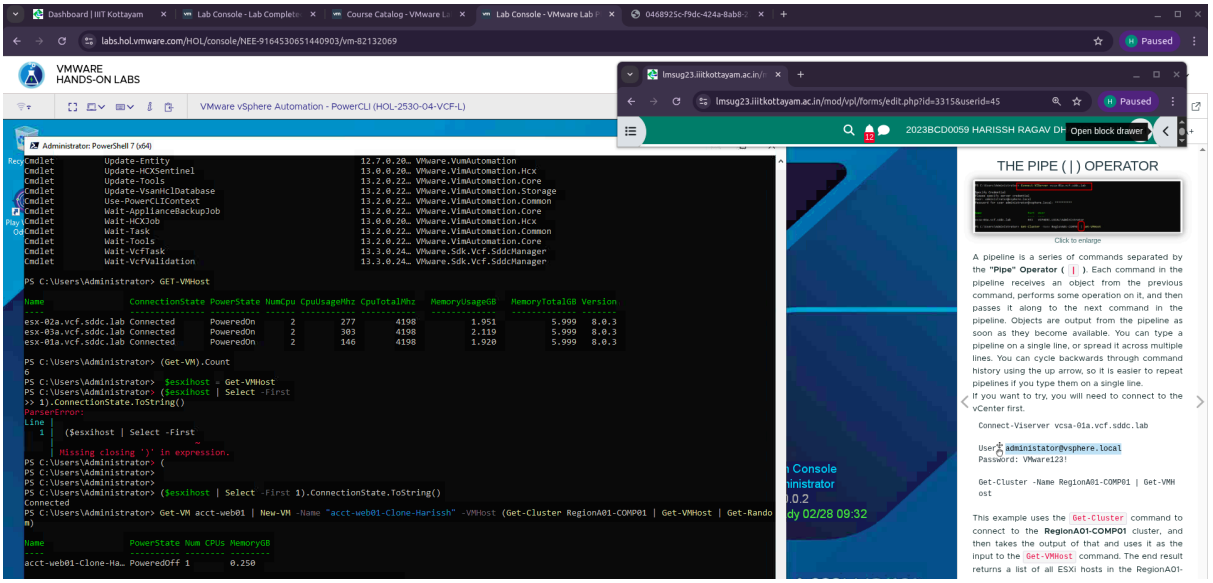
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Line |
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PS C:\Users\Administrator>
PS C:\Users\Administrator>
PS C:\Users\Administrator> ($esxihost | Select -First 1).ConnectionState.ToString()
Connected
PS C:\Users\Administrator> Get-VM acct-web01 | New-VM -Name "acct-web01-Clone-Harissh" -VMHost (Get-Cluster RegionA01-COMP01 | Get-VMHost | Get-Rando
m)
Name PowerState Num CPUs MemoryGB
acct-web01-Clone-Ha… PoweredOff 1 0.250
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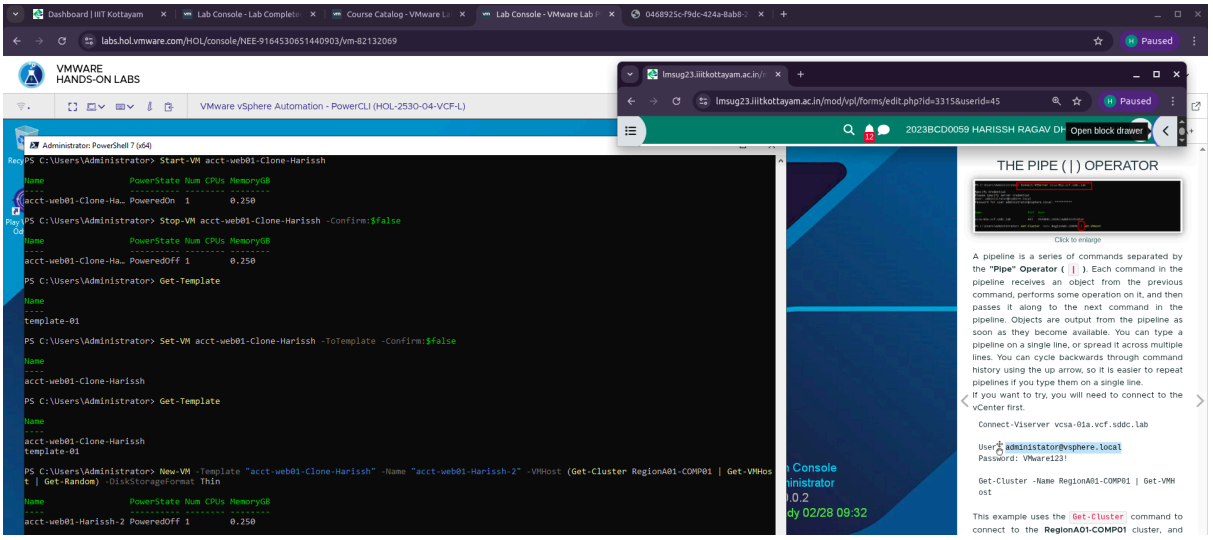
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labs.hol.vmware.com/HOL/console/NEE-9164530651440903/vm-82132069
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lmsug23.iiitkottayam.ac.in/mod/vpl/forms/edit.php?id=3315&userid=45
2023BCD0059 HARISSH RAGAV DH
Open block drawer
Administrator: PowerShell 7 (x64)
PS C:\Users\Administrator> Start-VM acct-web01-Clone-Harissh
Name PowerState Num CPUs MemoryGB
acct-web01-Clone-Ha… PoweredOn 1 0.250
PS C:\Users\Administrator> Stop-VM acct-web01-Clone-Harissh -Confirm:$false
Name PowerState Num CPUs MemoryGB
acct-web01-Clone-Ha… PoweredOff 1 0.250
PS C:\Users\Administrator> Get-Template
Name
template-01
PS C:\Users\Administrator> Set-VM acct-web01-Clone-Harissh -ToTemplate -Confirm:$false
Name
acct-web01-Clone-Harissh
PS C:\Users\Administrator> Get-Template
Name
acct-web01-Clone-Harissh
template-01
PS C:\Users\Administrator> New-VM -Template "acct-web01-Clone-Harissh" -Name "acct-web01-Harissh-2" -VMHost (Get-Cluster RegionA01-COMP01 | Get-VMHos
t | Get-Random) -DiskStorageFormat Thin
Name PowerState Num CPUs MemoryGB
acct-web01-Harissh-2 PoweredOff 1 0.250
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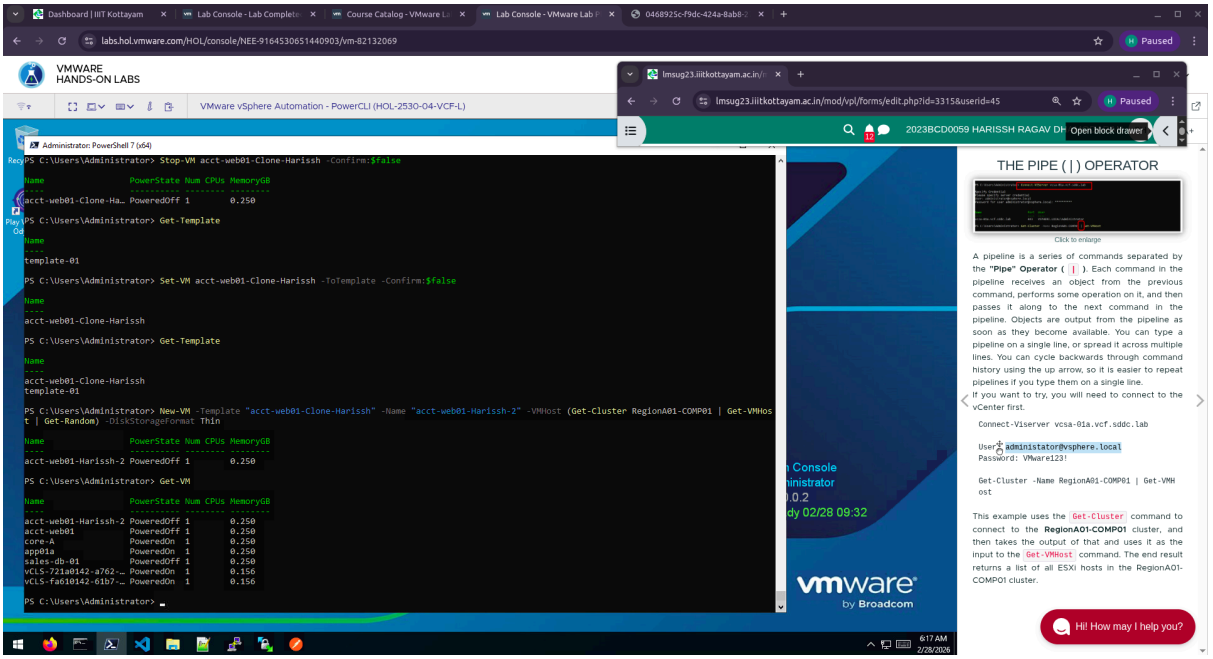
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lmsug23.iiitkottayam.ac.in/mod/vpl/forms/edit.php?id=3315&userid=45
2023BCD0059 HARISSH RAGAV DH
Open block drawer
Administrator: PowerShell 7 (x64)
PS C:\Users\Administrator> Stop-VM acct-web01-Clone-Harissh -Confirm:$false
Name PowerState Num CPUs MemoryGB
---- ---------- -------- --------
acct-web01-Clone-Ha… PoweredOff 1 0.250
PS C:\Users\Administrator> Get-Template
Name
---
template-01
PS C:\Users\Administrator> Set-VM acct-web01-Clone-Harissh -ToTemplate -Confirm:$false
Name
---
acct-web01-Clone-Harissh
PS C:\Users\Administrator> Get-Template
Name
---
acct-web01-Clone-Harissh
template-01
PS C:\Users\Administrator> New-VM -Template "acct-web01-Clone-Harissh" -Name "acct-web01-Harissh-2" -VMHost (Get-Cluster RegionA01-COMP01 | Get-VMHost | Get-Random) -DiskStorageFormat Thin
Name PowerState Num CPUs MemoryGB
---- ---------- -------- --------
acct-web01-Harissh-2 PoweredOff 1 0.250
PS C:\Users\Administrator> Get-VM
Name PowerState Num CPUs MemoryGB
---- ---------- -------- --------
acct-web01-Harissh-2 PoweredOff 1 0.250
acct-web01 PoweredOff 1 0.250
core-A PoweredOn 1 0.250
app01a PoweredOn 1 0.250
sales-db-01 PoweredOff 1 0.250
vCLS-721a0142-a762-… PoweredOn 1 0.156
vCLS-fa610142-61b7-… PoweredOn 1 0.156
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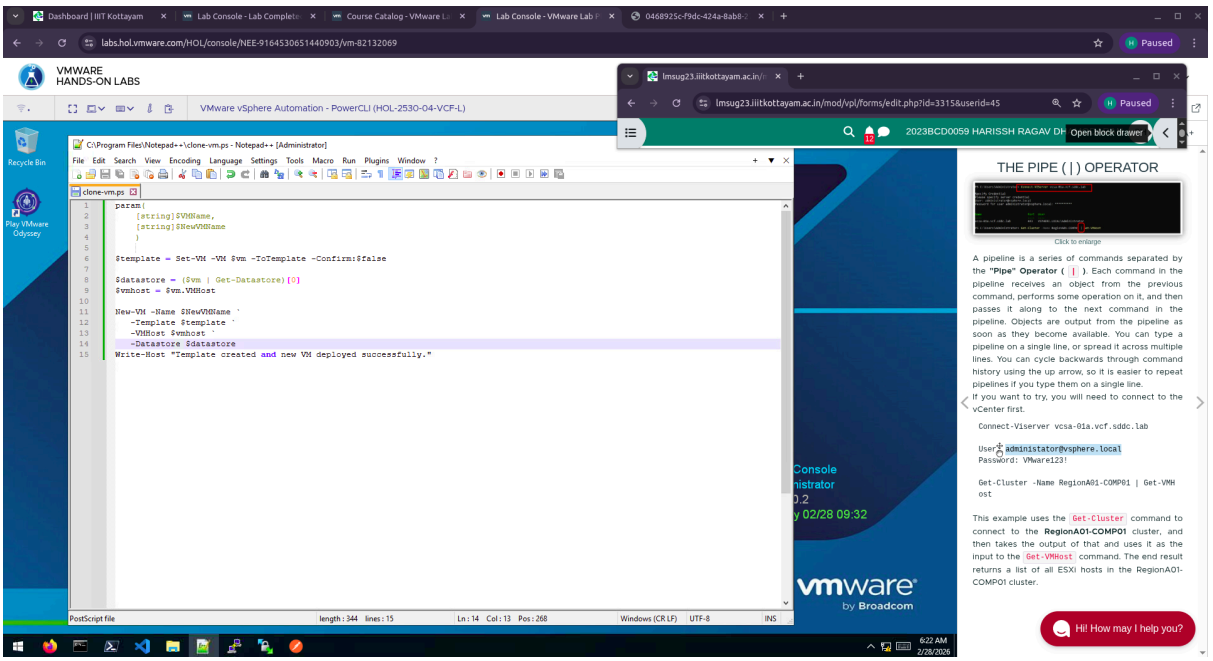
vmware by Broadcom
6:17 AM
2/28/2026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2023BCD0059 HARISSH RAGAV DH
Open block drawer
Recycle Bin
Play VMware Odyssey
C:\Program Files\Notepad++\clone-vm.ps - Notepad++ [Administrator]
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
clone-vm.ps
param(
[string]$VMName,
[string]$NewVMName
)
$template = Set-VM -VM $vm -ToTemplate -Confirm:$false
$datastore = ($vm | Get-Datastore)[0]
$vmhost = $vm.VMHost
New-VM -Name $NewVMName `
-Template $template `
-VMHost $vmhost `
-Datastore $datastore
Write-Host "Template created and new VM deployed successfully."
PostScript file
length : 344 lines : 15
Ln : 14 Col : 13 Pos : 268
Windows (CR LF)
UTF-8
INS
THE PIPE (|) OPERATOR
Click to enlarge
A pipeline is a series of commands separated by the "Pipe" Operator (|). Each command in the pipeline receives an object from the previous command, performs some operation on it, and then passes it along to the next command in the pipeline. Objects are output from the pipeline as soon as they become available. You can type a pipeline on a single line, or spread it across multiple lines. You can cycle backwards through command history using the up arrow, so it is easier to repeat pipelines if you type them on a single line.
If you want to try, you will need to connect to the vCenter first.
Connect-Viserver vcsa-01a.vcf.sddc.lab
User: administator@vsphere.local
Password: VMware123!
Get-Cluster -Name RegionA01-COMP01 | Get-VMHost
This example uses the Get-Cluster command to connect to the RegionA01-COMP01 cluster, and then takes the output of that and uses it as the input to the Get-VMHost command. The end result returns a list of all ESXi hosts in the RegionA01-COMP01 cluster.
Hi! How may I help you?
vmware by Broadcom
6:22 AM
2/28/2026